

BarchScan — Business Plan

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1. Executive Summary

BarchScan is a web-based SaaS platform — an **Intelligent Records Platform** — that transforms physical structured records of any kind into searchable, analyzable, encrypted digital data. Users photograph any paper record; an AI vision model extracts the tabular data; users review and edit the suggestion; and the confirmed data is encrypted and stored securely in the cloud.

BarchScan is broader than a logbook tool. It serves any organization that generates paper-based structured records — patrol sheets, inspection forms, attendance registers, inventory counts, compliance sheets, shift logs, lab notebooks, and more.

The platform operates on a **two-tier subscription model**:

Tier	Name	Monthly Price	Core Value
Tier 1	Records Vault	\$8/month	Scan, digitize, encrypt, store, search, and export any physical records
Tier 2	Intelligence Engine	\$23/month	Everything in Tier 1 + AI analytics assistant, manual analytics tools, external data upload, MCP-embedded tools, and deep data insights

BarchScan targets organizations that still depend on paper records — maritime operators, security firms, healthcare facilities, education institutions, manufacturing plants, agricultural operations, and government agencies — providing a path from paper to intelligence without manual data entry overhead.

2. Problem Statement

2.1 The Paper Records Bottleneck

Millions of organizations worldwide maintain physical records for operations, inspections, attendance, compliance, and incident tracking. These records are:

- **Inaccessible** — Retrieving a specific entry requires physically searching through pages.
- **Fragile** — Paper degrades, gets lost, or is destroyed by water, fire, or wear.
- **Unanalyzable** — Trends, anomalies, and patterns buried in handwritten data are invisible.
- **Non-compliant** — Regulatory audits demand searchable, timestamped, encrypted digital records.
- **Siloed** — Data in physical records cannot be combined with digital data for analysis.

2.2 Why Existing Solutions Fall Short

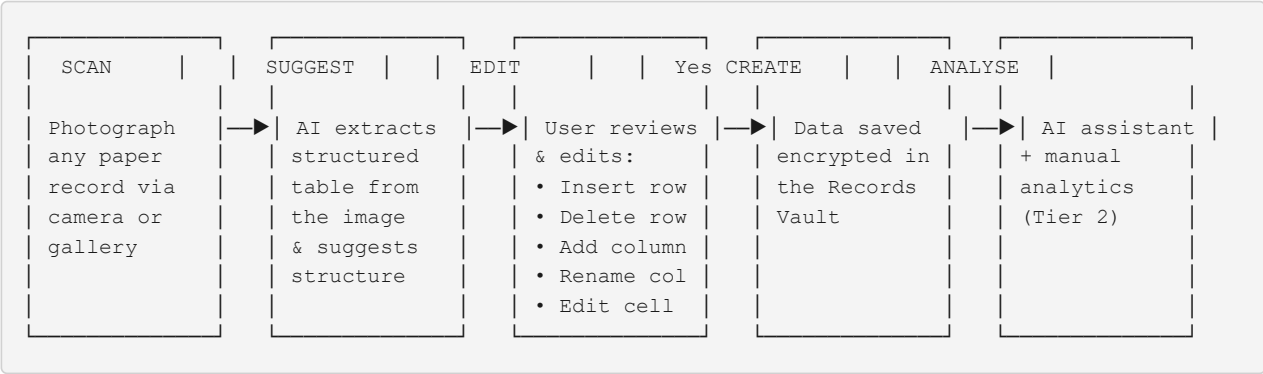
Existing Approach	Why It Fails
Manual data entry	Slow, expensive, error-prone. A full-time data clerk costs far more than \$8/month.
Generic OCR tools	Cannot interpret tabular layouts, maintain schema consistency, or handle handwriting.
Spreadsheet software	Requires manual entry. No image input. No intelligence.
Enterprise digitization suites	\$500+/month. Priced out of reach for most organizations. No AI analytics.
Consumer scan apps	Produce image files, not structured data. No analytics. No encryption.

2.3 The Opportunity

No affordable, intelligent, mobile-first platform currently transforms paper records into encrypted, analyzable data and connects users to an AI that can reason over that data. BarchScan fills this gap entirely.

3. Solution

BarchScan delivers a seamless pipeline from physical record to digital intelligence:



Key differentiators:

- 1. **AI-guided, human-confirmed** — Not blind OCR. Users review the AI suggestion and control what gets saved.
- 2. **Beyond logbooks** — BarchScan handles any structured physical record from any sector.
- 3. **Encrypted by default** — AES-256 encryption at rest, TLS in transit. Data privacy is non-negotiable.
- 4. **Connected intelligence** — Tier 2 AI assistant has access to all user records AND uploaded external data simultaneously, with tagging and chat referencing.
- 5. **External data upload** — Tier 2 users bring their own CSV/Excel/JSON files into the analytics engine alongside scanned records.

4. Product Overview

4.1 Tier 1 — Records Vault (\$8/month)

The Records Vault digitizes, encrypts, and organizes any physical structured record for easy retrieval and documentation.

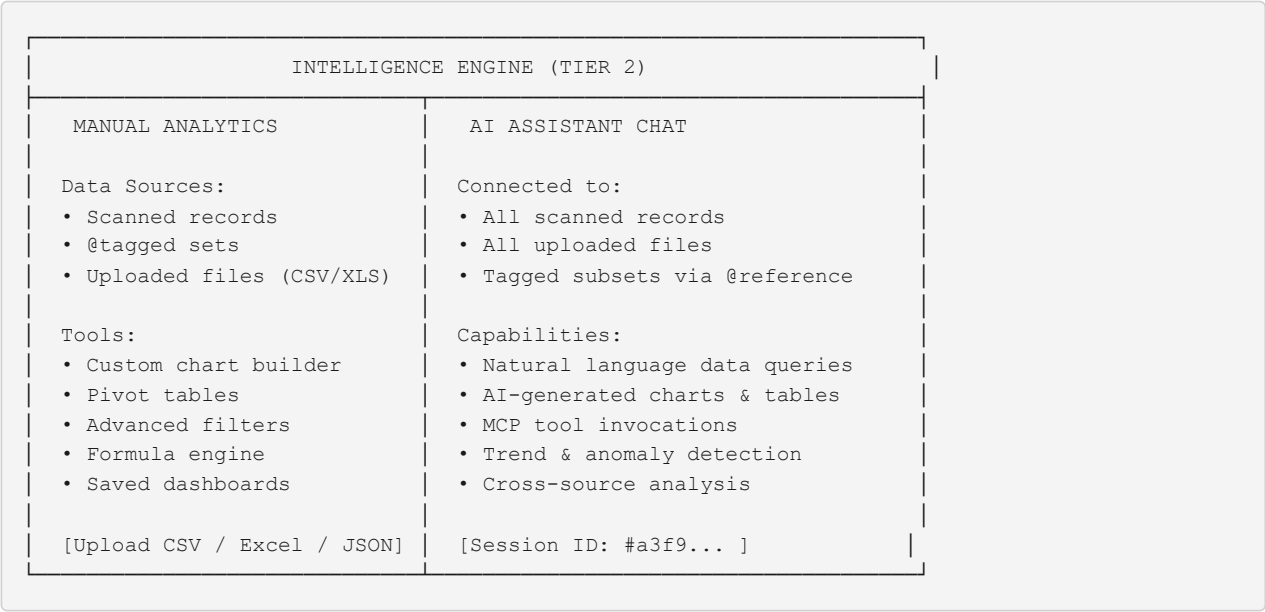
Feature	Description
Scan & Extract	Photograph any record page; AI extracts structured tabular data
AI Suggestion Review	Review the AI's table interpretation before committing any data
Table Editor	Insert/delete rows, add/remove columns, rename headers, edit cells — full control
Create & Save	Confirm edits and save to AES-256 encrypted cloud storage
Records Retrieval	Search, filter, sort, and browse all digitized records
Export	Download as CSV, JSON, or copy to clipboard
Encryption	AES-256-GCM at rest + TLS 1.3 in transit
User Accounts	Individual accounts with Supabase Auth (email + OAuth)
Session Management	Secure session management, session isolation, and support traceability
Multi-device Access	Access all records from any device via web browser
Image Archive	Original scanned images stored alongside extracted data in Cloudinary

4.2 Tier 2 — Intelligence Engine (\$23/month)

Everything in Tier 1, plus a world-class analytics and AI layer. Designed to grow from basic analysis to complex, production-grade data intelligence.

Feature	Description
AI Analytics Assistant	Embedded AI (chat panel) with access to all user records and uploaded data
Tag & Reference System	Tag record sets or datasets; reference them in AI chat via <code>@tagname</code>
External Data Upload	Upload CSV, Excel, or JSON files into the analytics engine for AI analysis and cross-referencing
Automated Charts & Graphs	AI generates contextually appropriate visualizations on demand
MCP-Embedded Tools	AI uses embedded tools to sort, filter, group, and render data within the chat thread
Manual Analytics	Custom chart builder, pivot tables, advanced filters, formula engine, saved dashboards
Deep Insights	Trend detection, anomaly detection, pattern recognition, correlation analysis

4.2.1 Intelligence Engine Architecture



5. Business Model

5.1 Pricing Structure

	Tier 1 — Records Vault	Tier 2 — Intelligence Engine
Monthly Price	\$8	\$23
Annual Price	\$76.80 (20% off)	\$220.80 (20% off)
Target User	Officers, clerks, operations managers	Analysts, compliance officers, decision-makers
Core Value	"Never lose a record again"	"Turn your records into intelligence"

5.2 Free Tier (Growth Funnel)

Aspect	Free Tier
Scans per month	10
Records stored	Up to 100 rows
Analytics	None
Encryption	Yes Same as paid
Purpose	Trial → conversion to Tier 1 or Tier 2

5.3 Revenue Formula

$$\text{MRR} = (\text{Tier 1 users} \times \$8) + (\text{Tier 2 users} \times \$23)$$
$$\text{ARR} = \text{MRR} \times 12$$

5.4 Unit Economics

Metric	Tier 1	Tier 2
Monthly price	\$8	\$23
Est. COGS/user/month (infra + AI API)	~\$1.50	~\$5.00
Gross margin	~81%	~78%
12-month LTV	\$96	\$276
Target CAC	<\$30	<\$70
LTV:CAC ratio	>3:1	>4:1

6. Market Analysis

6.1 Target Segments

Segment	Use Case	Pain Point	Tier Fit
Maritime / Shipping	Vessel logs, port inspection records, cargo manifests	Paper compliance is hard to audit	Tier 1 → Tier 2
Security / Guard Services	Patrol logs, incident reports, visitor registers	Handwritten entries are unsearchable	Tier 1
Healthcare / Clinics	Patient check-in logs, medication administration records	Regulatory compliance demands digital records	Tier 1 → Tier 2
Education	Attendance registers, lab notebooks, exam records	Schools need digital backups and admin analytics	Tier 1
Manufacturing	Shift logs, equipment inspection records, quality checks	Downtime and quality analysis requires digitized shift data	Tier 2
Government / Public Sector	Registry offices, immigration logs, public records	Transparency and searchability mandates	Tier 1 → Tier 2
Agriculture	Farm records, harvest logs, spray and irrigation schedules	Traceability requirements for certification	Tier 1
Small Businesses	Visitor logs, inventory counts, delivery records	No budget for enterprise digitization	Tier 1

6.2 Market Sizing

Metric	Estimate	Basis
TAM	~\$4.2B/year	Global document digitization + data analytics market for SMBs
SAM	~\$320M/year	Organizations actively using physical structured records in English-speaking markets
SOM	~\$3.2M/year	Conservative 1% penetration in Year 1–2 (15,000–20,000 paying users)

6.3 Market Tailwinds

- **Regulatory pressure:** GDPR, HIPAA, maritime MLC, and sector-specific mandates drive digital record-keeping requirements worldwide.
- **Mobile-first emerging markets:** Organizations in fast-growing economies are leapfrogging desktop software — phone camera is the natural input device.
- **AI democratization:** Gemini and similar models bring intelligent extraction to commodity pricing.
- **Encryption demand:** Data privacy awareness post-2020 drives demand for encrypted-by-default platforms.

- **Cross-data analytics:** Organizations increasingly need to combine operational records with other data sources — exactly what Tier 2 delivers.

7. Competitive Landscape

7.1 Competitive Matrix

Competitor	Price	AI Extraction	Human Review Step	Encrypted Storage	Analytics	External Data Upload	Mobile-First
BarchScan	\$8–\$23/mo	Yes AI Vision	Yes Full editor	Yes AES-256	Yes AI + Manual	Yes CSV/Excel/JSON	Yes
Google Lens	Free	Partial OCR	No	No	No	No	Yes
Adobe Scan	\$10–\$25/mo	OCR only	No	No	No	No	Yes
Microsoft Lens	Free	Basic table	Limited	No	No	No	Yes
Tabula	Free	Rule-based	No	No	No	No	No
ABBYY FineReader	\$200+/mo	Yes	Limited	Yes	No	No	No
Kofax	\$500+/mo	Yes	Yes	Yes	Limited	No	No

7.2 BarchScan's Moat

1. **Price** — At \$8/month, BarchScan is 5–60× cheaper than the only paid alternatives with comparable extraction quality.
 2. **Human-in-the-loop** — No other consumer/SMB tool has an AI suggestion + user edit + confirm workflow.
 3. **"Records" positioning** — Not locked to logbooks. Serves any sector with structured paper records.
 4. **Connected analytics** — Tier 2 connects scanned records AND uploaded external data to a single AI assistant. No competitor does this.
 5. **Encryption by default** — Data privacy is built in, not an enterprise add-on.
 6. **Mobile-first** — Designed for the phone camera. No scanner hardware required.
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8. Go-to-Market Strategy

8.1 Launch Phases

Phase 1 — Foundation (Months 1–3)

- Launch Tier 1 with user accounts, encryption, and the scan-suggest-edit-create workflow.
- Beta with 50–100 users across maritime, security, and healthcare sectors.
- Gather feedback, iterate on the Table Editor UX and mobile experience.
- Integrate Stripe billing and tier enforcement.

Phase 2 — Growth (Months 4–6)

- Launch Tier 2: manual analytics (chart builder, pivot tables, filters).
- AI analytics assistant v1: chat interface, preset charts, @tag referencing.
- External data upload (CSV, Excel, JSON) for Tier 2 users.
- Content marketing: blog posts, video demos, vertical case studies.

Phase 3 — Scale (Months 7–12)

- MCP-embedded tools in AI assistant — fully agentic data operations.
- Advanced analytics: trend detection, anomaly detection, statistical analysis.
- Annual billing with 20% discount.
- Partner program with maritime, healthcare, and security industry associations.
- Multi-language extraction support.

Phase 4 — Enterprise (Year 2+)

- Team accounts with admin controls and audit trails.
- API access for integrations with ERP, CRM, and compliance systems.
- White-label option for large organizations.
- SOC 2 Type II certification.

8.2 Acquisition Channels

Channel	Strategy	Expected CAC
SEO	Target "digitize records", "scan handwritten forms", "paper to spreadsheet AI"	\$5–10
Content Marketing	Vertical case studies (maritime, security), ROI calculators, tutorials	\$10–15
Social / Video	LinkedIn B2B outreach; demo videos on TikTok and YouTube	\$15–25
Industry Partnerships	Associations, trade bodies, sector-specific distributors	\$20–30
Referral Program	1 month free for each referred paying user	\$8–23
Free → Paid Funnel	In-app upgrade prompts when free limits are reached	\$2–5

8.3 Retention Strategy

Strategy	Implementation
Data lock-in (positive)	The more records stored, the more valuable the vault and analytics become
Analytics upgrades	Continuous Tier 2 feature additions drive ongoing engagement and reduce churn
Weekly digests	Automated email summaries of records added and analytics highlights
Tier 1 → Tier 2 upsell	In-app prompts: "You have 500 records. Unlock trends and insights with Intelligence Engine."

9. Revenue Projections

9.1 Conservative Scenario (Year 1)

Quarter	Tier 1 Users	Tier 2 Users	MRR	Quarterly Revenue
Q1	100	20	\$1,260	\$3,780
Q2	350	70	\$4,410	\$13,230
Q3	800	180	\$10,540	\$31,620
Q4	1,500	400	\$21,200	\$63,600
Year 1 Total				\$112,230

9.2 Moderate Scenario (Year 1)

Quarter	Tier 1 Users	Tier 2 Users	MRR	Quarterly Revenue
Q1	250	50	\$3,150	\$9,450
Q2	800	180	\$10,540	\$31,620
Q3	2,000	500	\$27,500	\$82,500
Q4	4,000	1,200	\$59,600	\$178,800
Year 1 Total				\$302,370

9.3 Cost Structure (Monthly at Scale)

Cost Category	Monthly Estimate	Notes
Gemini API	\$500–\$2,000	Usage-based; scales with scans and analytics sessions
Supabase	\$200–\$800	Scales with data volume and active users
Cloudinary	\$50–\$200	Image storage and CDN
Stripe fees	2.9% of revenue	Standard payment processing
Vercel	\$20–\$200	Serverless compute and CDN
Team	Variable	Founders + engineering + growth
Marketing	\$500–\$2,000	Content, ads, partnerships

10. Technology & Infrastructure

10.1 Current Stack (v1.0 — Live)

Layer	Technology
Frontend	Vanilla HTML/CSS/JS
Backend	Vercel Serverless (Node.js)
AI	Google Gemini 2.5 Flash
Database	JSONBin.io (being migrated)
Image Store	Cloudinary

10.2 Target Stack (v2.0 — SaaS Platform)

Layer	Technology	Purpose
Frontend	HTML/CSS/JS → component-based (analytics)	Scale for analytics complexity
Backend	Vercel Serverless (Node.js)	Zero-config, auto-scaling
AI	Gemini 2.5 Flash (vision + function calling)	Extraction + analytics assistant
Database	Supabase (PostgreSQL + RLS + Auth)	Multi-tenant, encrypted, scalable
Payments	Stripe Billing	Subscription and tier management
Encryption	AES-256-GCM (app-level) + Supabase at-rest	Defense in depth
MCP Tools	Model Context Protocol	Agentic AI tool use in analytics

10.3 Scaling Path

Phase	Users	Infrastructure
Phase 1	0–1K	Vercel Hobby/Pro · Supabase Free · Stripe Basic
Phase 2	1K–10K	Vercel Pro · Supabase Pro · Cloudinary Plus
Phase 3	10K+	Vercel Enterprise · Supabase Enterprise · Custom infra

11. Security & Compliance

11.1 Encryption Architecture

Layer	Mechanism
Transport	TLS 1.3 — all client-server communication
Application	AES-256-GCM — per-record encryption before database write
Storage	Supabase at-rest encryption (AES-256)
Sessions	AES-256-GCM — analytics session context encrypted per session ID

11.2 Data Privacy Commitments

Commitment	Details
User data ownership	Users own 100% of their data. BarchScan is a processor, not an owner.
No data selling	User data is never sold, shared, or used for advertising.
Right to export	Users can export all data at any time (CSV, JSON).
Right to delete	Users can permanently delete all records and close their account.
AI isolation	Data sent to Gemini for analytics is not stored for model training.
Session isolation	Each analytics session is completely isolated between users.

11.3 Compliance Roadmap

Standard	Timeline	Why It Matters
GDPR compliance	Phase 1	European users, data processing agreements
SOC 2 Type II	Year 2	Enterprise customer requirement
HIPAA compliance	Year 2	Healthcare logbook and patient record users
ISO 27001	Year 3	Enterprise and government credibility

12. Team & Roles

12.1 Roles Needed

Role	Responsibility	Priority
Product / Founder	Vision, strategy, user research, GTM execution	Immediate
Full-Stack Engineer	Frontend + backend, Vercel, Supabase, Stripe	Immediate
AI / ML Engineer	Analytics engine, Gemini function calling, MCP tools	Phase 2
Designer	UI/UX for Table Editor, analytics dashboard, mobile	Phase 1
Growth / Marketing	Content, SEO, partnerships, conversion optimization	Phase 2
Customer Success	Onboarding, retention, feedback loops	Phase 3

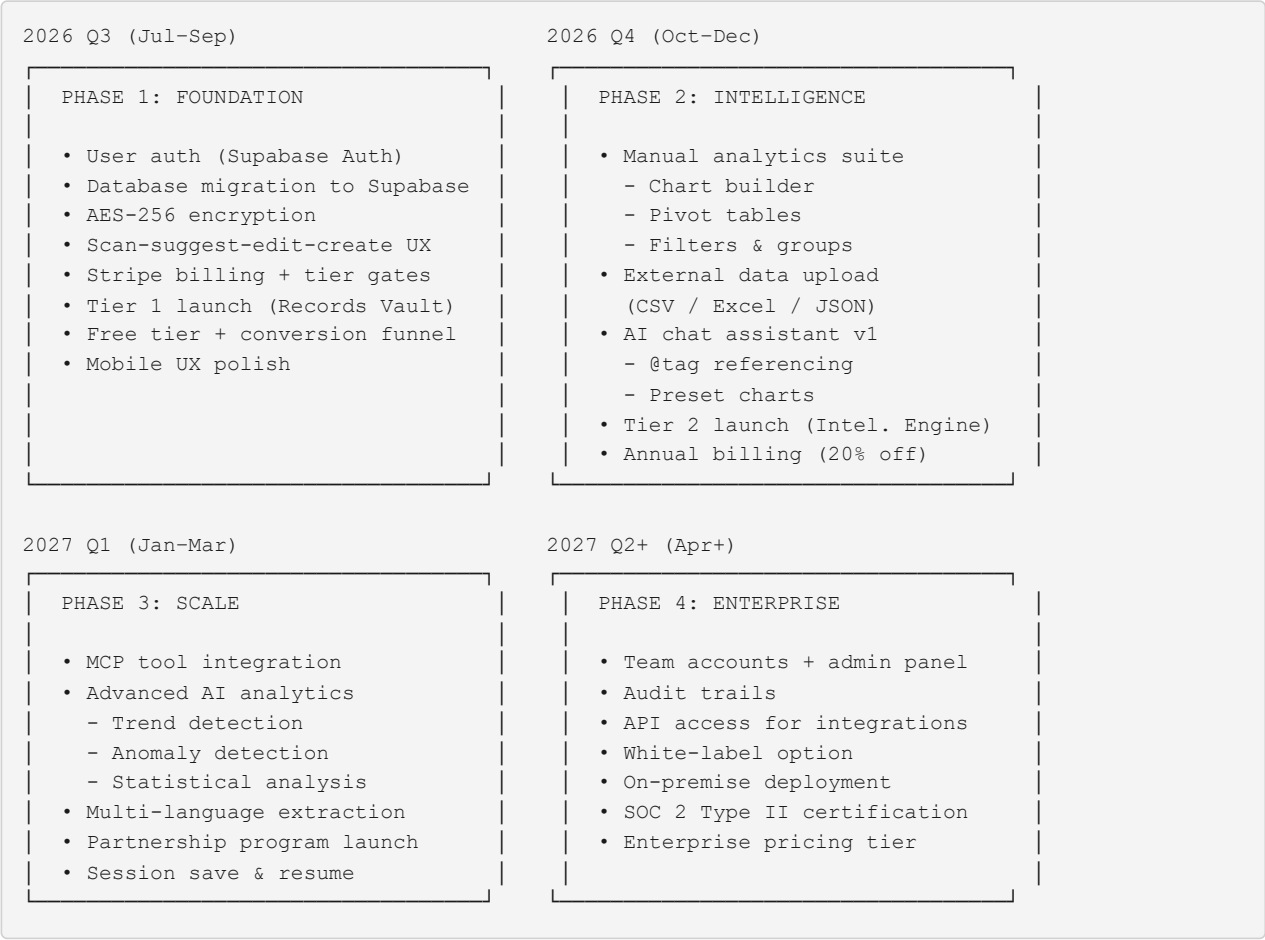
12.2 Hiring Plan

- **Phase 1 (Months 1–3):** Founder(s) + 1 full-stack engineer + 1 contract designer
- **Phase 2 (Months 4–6):** + 1 AI engineer + 1 growth hire
- **Phase 3 (Months 7–12):** + 1 additional engineer + 1 customer success manager

13. Risks & Mitigations

Risk	Severity	Likelihood	Mitigation
AI extraction quality on poor handwriting	High	Medium	Human-in-the-loop edit step; iterative prompt engineering
Competitor entry (Google/Microsoft adds features)	High	Low	Speed to market; vertical specialization; analytics moat; price
Gemini API cost escalation	Medium	Medium	Multi-model fallbacks (OpenAI, Claude); batch processing optimization
Encryption performance overhead	Medium	Medium	Efficient key management; selective encryption; edge caching
User adoption in paper-heavy industries	High	High	Free tier; ROI calculators; partner with associations; training materials
Churn after records fully digitized	Medium	Medium	Tier 2 ongoing analytics value; new record generation (ongoing ops); dashboards
Database migration complexity (JSONBin → Supabase)	Medium	Certain	Gradual migration; backward compatibility; thorough testing

14. Roadmap



14.1 Intelligence Engine Investment Areas

The Tier 2 analytics engine is the primary long-term investment. Goal: **world-class data intelligence**.

Investment Area	Description	Priority
AI Chat Interface	Natural language queries over all user data (records + uploads)	Critical
External Data Upload	CSV/Excel/JSON ingestion into analytics engine	Critical
Tag & Reference System	@tagname referencing in chat for any record set or dataset	Critical
MCP Tool Integration	AI invokes tools to sort, filter, render data in chat	Critical
Automated Visualization	AI generates contextually right charts without configuration	Critical
Session Infrastructure	Encrypted, managed, resumable session IDs	Critical
Manual Analytics	From basic (bar charts) to complex (regression, correlation)	High
Export & Reporting	PDF reports, scheduled email digests of insights	Medium
Custom Dashboards	User-saved analytics views	Medium

15. Key Metrics

15.1 North Star Metrics

Metric	Description	Year 1 Target
MRR	Monthly Recurring Revenue	\$20K+ by Q4
Active Users	Users who scanned or accessed data in last 30 days	2,000+
Scans/User/Month	Average scans per active user per month	8+

15.2 Growth Metrics

Metric	Target
New signups/month	500+ by Q4
Free → Paid conversion	15%+
Tier 1 → Tier 2 upgrade	25%+
Monthly churn	<5%

15.3 Product Quality Metrics

Metric	Target
Extraction accuracy (cells correct)	90%+
Edit rate (user changes after AI suggestion)	<20%
Time from signup to first scan	<3 minutes
Analytics adoption (Tier 2 weekly users)	60%+

15.4 Financial Health

Metric	Target
Gross margin	>75%
LTV:CAC ratio	>3:1
Payback period	<4 months

End of Document — BarchScan Business Plan v2.0